Frequency Shifter

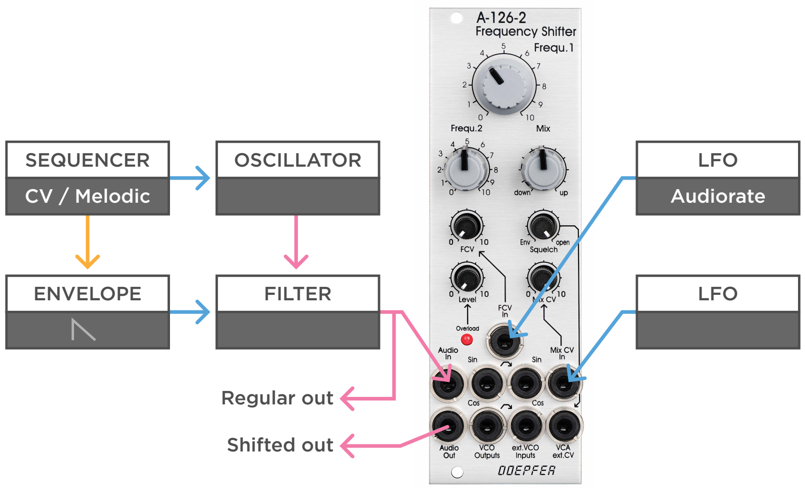
Synth & Drone Patches

Audio

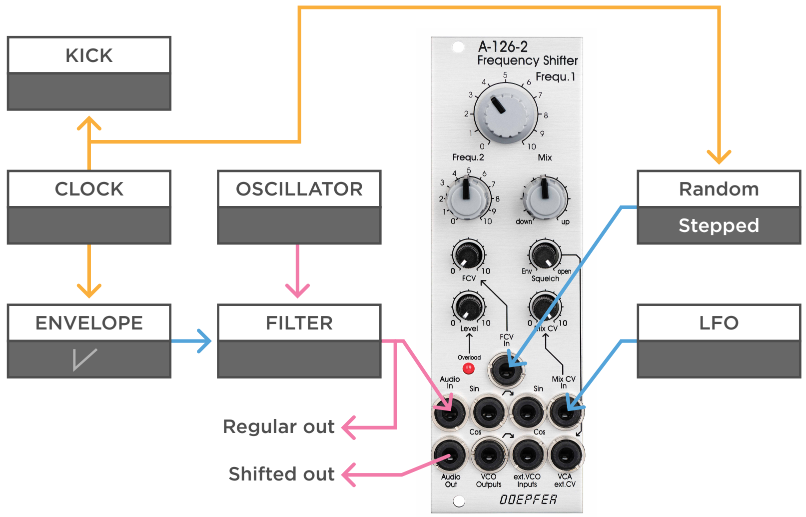
Control Voltage

Trigger / Gate

MONOTRAIL TECH TALK



(a)



(b)

a - Let's start with a simple mono synth voice, made with an oscillator, filter, sequencer and envelope. Run the voice through the frequency shifter, and send both a copy of the clean signal , as well as the shifted version to an outboard mixer. This way you can use the frequency shifter to add texture to the original sound. You can add a bit of audio rate modulation to the Frequency, and a slow LFO to the MIX CV, to add some movement to the patch.

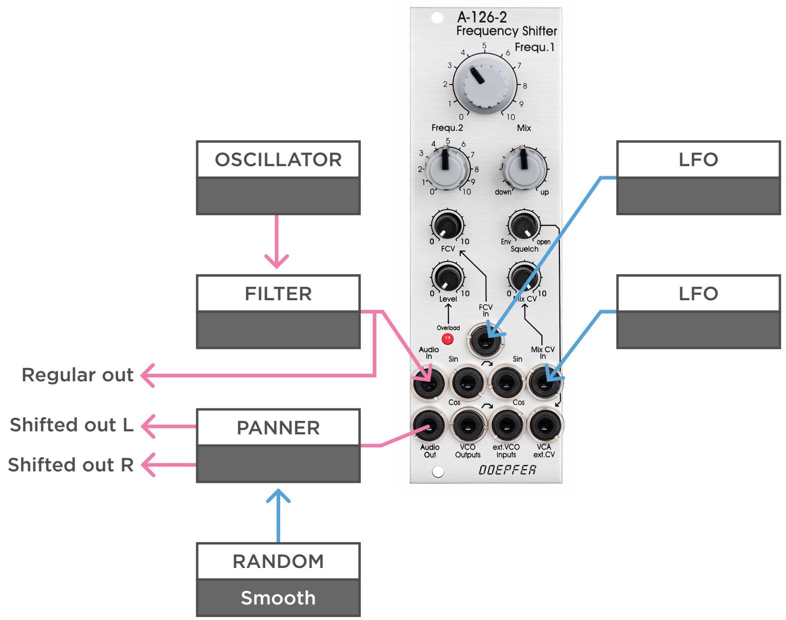
b - The frequency shifter also works great to add texture and depth to drones. Use a single oscillator through a filter to create a constant drone. Then, add a slow LFO to the MIX CV again, but this time, use a stepped random voltage to the Frequency input.

You can expand on this patch by adding a clock and use it to drive a steady kick, clock the stepped random voltage, and finally, trigger an inverted envelope. You can use the envelope to duck the filter a bit every time the kick hits. Now you have an interesting rhythmic patch to explore.

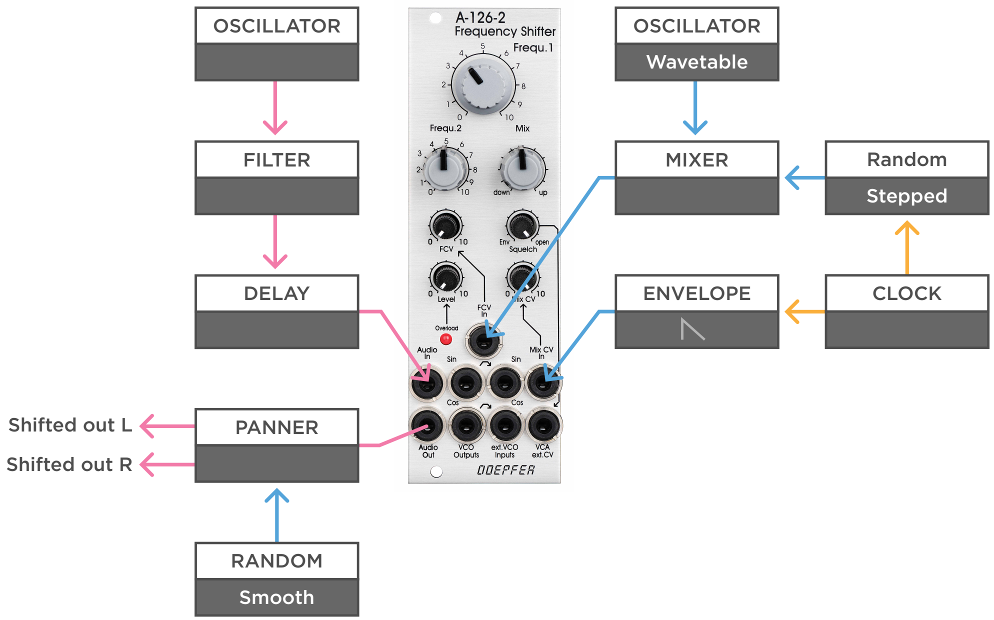
Frequency Shifter

Synth & Drone Patches

- Audio
- Control Voltage
- Trigger / Gate



(a)



(b)

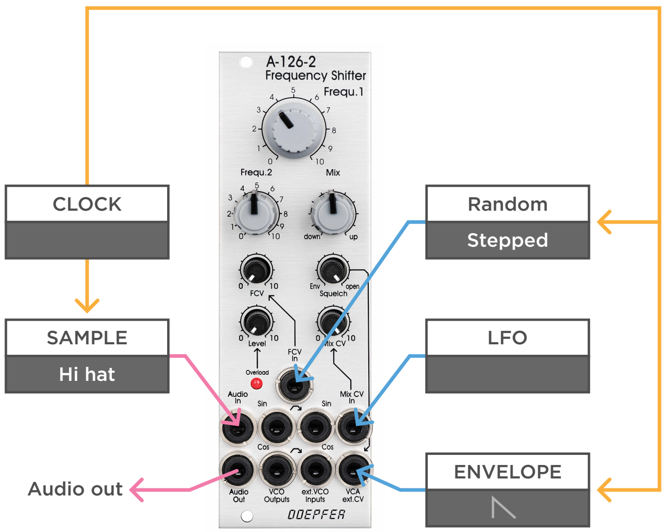
a - Another drone patch with a single static oscillator through a filter. This time with two slow LFOs with different frequencies modulating the Frequency and MIX CV. Beside listening to the clean output, send the frequency shifted version through a panner and use something like a smooth random voltage to spread the created textures around the stereo field. Reverb responds lovely to these complex harmonics.

b - You can expand the drone by adding a fast delay to add more complex harmonics. Then use a mixer to mix an audio rate signal from a wavetable oscillator and a stepped random voltage together, and send the mix to the Frequency input. Add a plucky envelope to the MIX CV, sweeping from an up, to a down shifted version, creating a bell like effect. And finally use a simple clock to trigger both the envelope, and set a new value for the stepped random voltage. In this case just listen to the frequency shifted output through a panner.

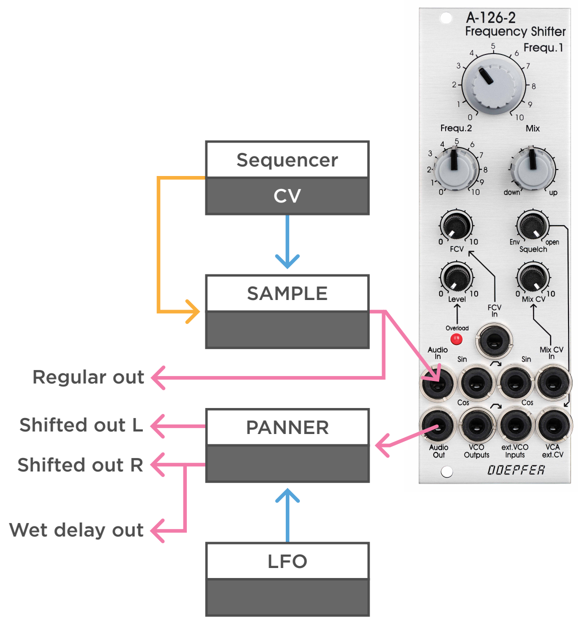
Frequency Shifter

Percussive Patches

Audio
Control Voltage
Trigger / Gate



(a)



(b)

a - A Frequency shifter also works great to add variations to percussive sounds, or create them from scratch. For example, you can easily add dynamics to a static sample, like a hi-hat triggered by a clock, by sending a tempo synced stepped random voltage to the Frequency input. You can use the squelch control and set it to the internal envelope follower if you feed the module short samples. But you can also trigger an external envelope and send it to the VCA input to control the length of the sound manually. Instead of modulating the Frequency, modulating just the MIX CV with something like an LFO can create more subtle movement.

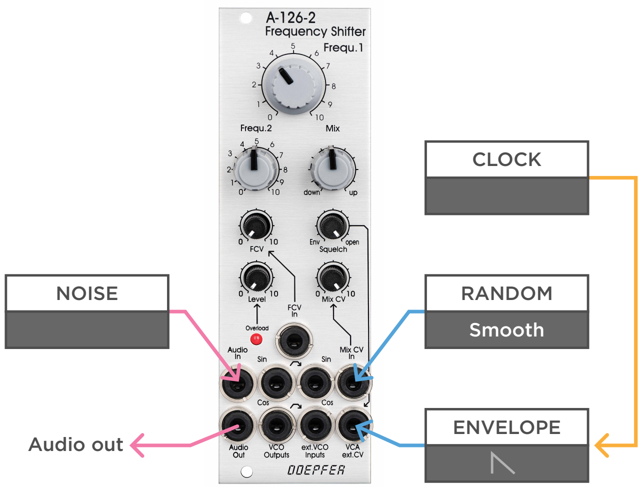
b - Again, stereo tricks works really well. In this patch a sequencer is used to sequence drum samples. A clean version is sent to an outboard mixer, and a copy to the pitch shifter. The output is sent through a panner again, with an LFO sweeping the sound left and right. The right signal is sent through a delay for some added texture.

Frequency Shifter

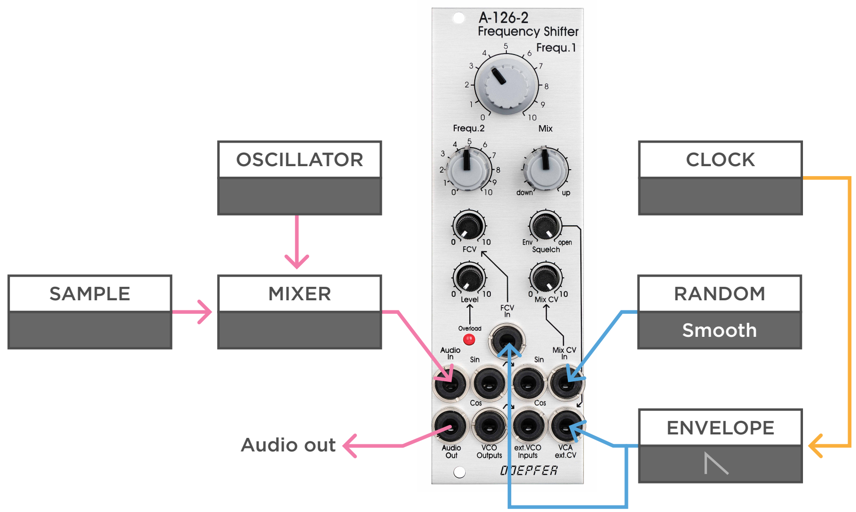
Percussive Patches

MONOTRAIL
TECH TALK

Audio
Control Voltage
Trigger / Gate



(a)



(b)

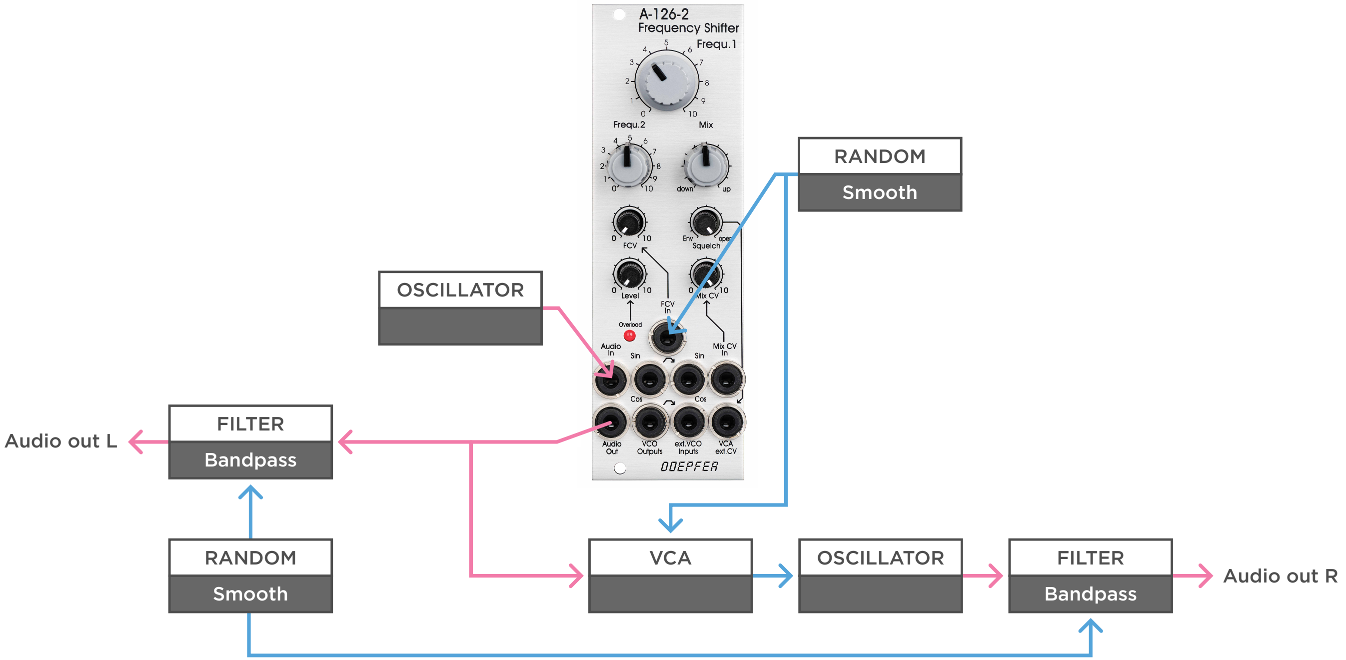
a - You can create hats yourself by feeding the module noise, and modulating the Frequency, or for more subtle effects the MIX CV. Send an envelope to the CV to create short tight sounds.

b - But of course you can feed the module anything you like to create percussive sounds. For example, a mix of a low tuned oscillator and a crackly field recording. Experiment by multiplying the envelope to the Frequency input, to create percussive sounds that go down in pitch.

Frequency Shifter

Bonus Patches

Audio
Control Voltage
Trigger / Gate



(a)

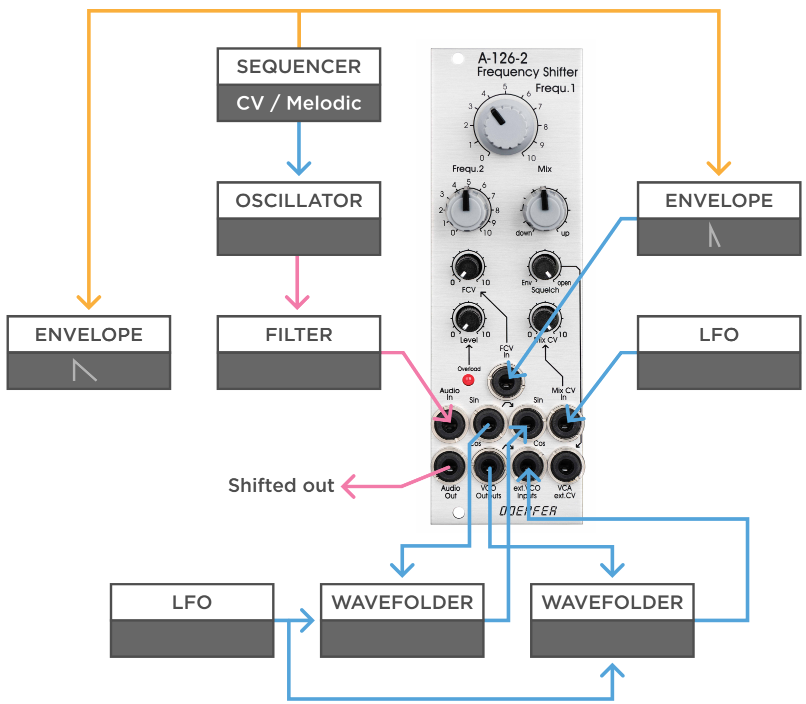
a - Frequency shifters work great in more complex self-generative FM patches. In this example two oscillators are used to create a stereo sound. One oscillator is passed through the frequency shifter, and then a band pass filter. The other oscillator also passes through a separate bandpass filter.

Then the output of the frequency shifter is multiplied, sent to a VCA, and passed on to modulate the frequency of the second oscillator. The modulation amount is modulated with a smooth random voltage, that at the same time, is modulating the Frequency of the shifter. Finally a second smooth random voltage is used to modulate the frequency of both bandpass filters, one with inversed polarity.

Frequency Shifter

Bonus Patches

Audio
Control Voltage
Trigger / Gate



(a)

a - In this example an oscillator and filter combo is used to create a simple voice, that is sent to the frequency shifter. Just listen to the direct output of this patch. A sequencer is used to tune the oscillator. It also triggers a short envelope opening the filter, as well as a very short envelope modulating the frequency, creating short wet transients to the sound. A slow LFO is used to modulate the MIX CV.

But now, take both the internal oscillator outputs, send them through wave folders, and back into the module. You can modulate the folders with something like a slow LFO. In this case one of the folders uses an attenuverter, to invert the incoming LFO. Experiments like this easily destroy the precise frequency shifting function, but at the same time, they can result both into more clean, as well as more noisy results.

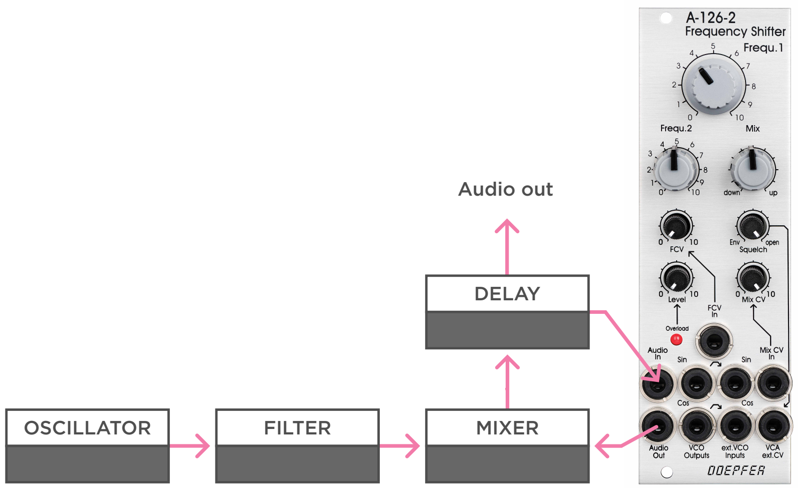
Frequency Shifter

Bonus Patches

Audio

Control Voltage

Trigger / Gate



(a)

(b)

a - It's a good idea to use a frequency shifter in effects or feedback loops. For example, use a simple synth voice. Send the voice to a mixer, and the mix output to a delay. Listen to the mix output of the delay, but send a wet output to the frequency shifter, and then back into the loop via the mixer. You can use the mixer to set the feedback amount, with every delay, passing through the shifter again.

b - Or use the frequency shifter in the feedback loop of a spring reverb. Ths creates lovely noisy soundscapes.